

Week 2 Cybersecurity Internship Report

Name: Lubanzi Fihla lubhilubanzi@gmail.com +27 78 973 2604

Batch: B083

Program: NetworkWalks Cybersecurity Internship

Disclaimer

This report was prepared for educational purposes as part of the NetworkWalks Cybersecurity Internship Program. All activities were performed in a controlled lab environment and publicly available information sources were used during reconnaissance activities.

Executive Summary

This report documents the completion of Week 2 cybersecurity activities involving footprinting using theHarvester and network scanning using Zenmap.

The objective was to understand how cybersecurity professionals perform reconnaissance and network discovery to identify publicly available information and active hosts within a network.

The practical exercises provided hands-on exposure to Open Source Intelligence (OSINT), host discovery, network enumeration, and use of industry-standard tools such as theHarvester, Zenmap, and Nmap.

Module 1: Footprinting with theHarvester

Objective

The objective of this activity was to gather publicly available information about a target domain using passive reconnaissance techniques.

Tool Used

- Kali Linux
- theHarvester

Commands Executed

theHarvester -d microsoft.com -l 1000 -b baidu

theHarvester -d microsoft.com -l 50 -b all

Results

TheHarvester successfully collected publicly available information related to the domain **microsoft.com**.

The scan identified numerous hostnames and subdomains associated with Microsoft services.

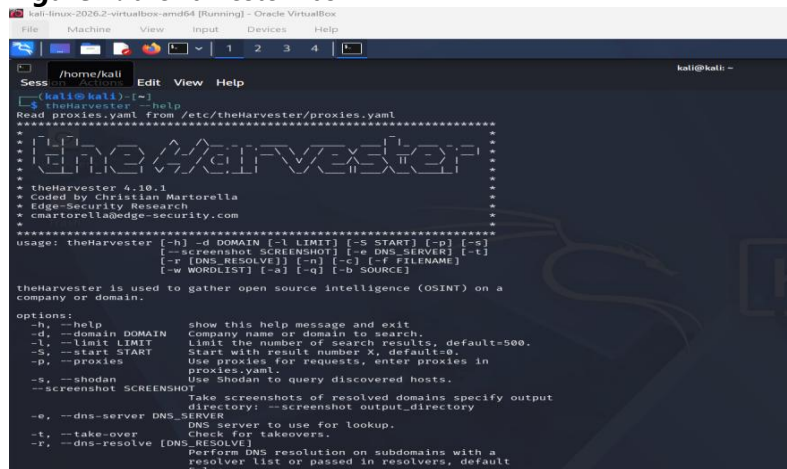
Examples discovered include:

- learn.microsoft.com
- support.microsoft.com
- developer.microsoft.com
- outlook.microsoft.com
- copilot.microsoft.com
- opensource.microsoft.com

The scan demonstrated how publicly available information can be collected without directly interacting with target systems.

Evidence

Figure 1: theHarvester Task 1



```
theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com

usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-s]
                  [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t]
                  [-r [DNS_RESOLVE]] [-a] [-c] [-f FILENAME]
                  [-w WORDLIST] [-a] [-q] [-b SOURCE]

theHarvester is used to gather open source intelligence (OSINT) on a
company or domain.

options:
  -h, --help            show this help message and exit
  -d, --domain DOMAIN    Company name or domain to search.
  -l, --limit LIMIT      Limit the number of search results, default-500.
  -s, --start START      Start with result number X, defaults 0.
  -p, --proxies           Use proxies for requests, enter proxies in
                        proxies.yaml.
  -s, --shodan           Use Shodan to query discovered hosts.
  --screenshot SCREENSHOT Take screenshots of resolved domains specify output
                        directory: --screenshot output_directory
  -e, --dns-server DNS_SERVER DNS server to use for lookup.
  -t, --take-over         Check for takeovers.
  -r, --dns-resolve [DNS_RESOLVE] Perform DNS resolution on subdomains with a
                        resolver list or passed in resolvers, default
                        False.
```

Figure 2: theHarvester Task 2

[illegible]

Module 2: Network Scanning with Zenmap

Objective

The objective of this activity was to identify active hosts within the virtual lab network.

Tool Used

- Zenmap
- Nmap
- Kali Linux

Network Details

Target Network:

10.0.0.0/24

Scan Profile:

Ping Scan

Scan Results

The Zenmap scan successfully identified active hosts on the network.

Live Hosts Discovered

10.0.0.1

10.0.0.2

Number of Active Hosts

2

MAC Address Identified

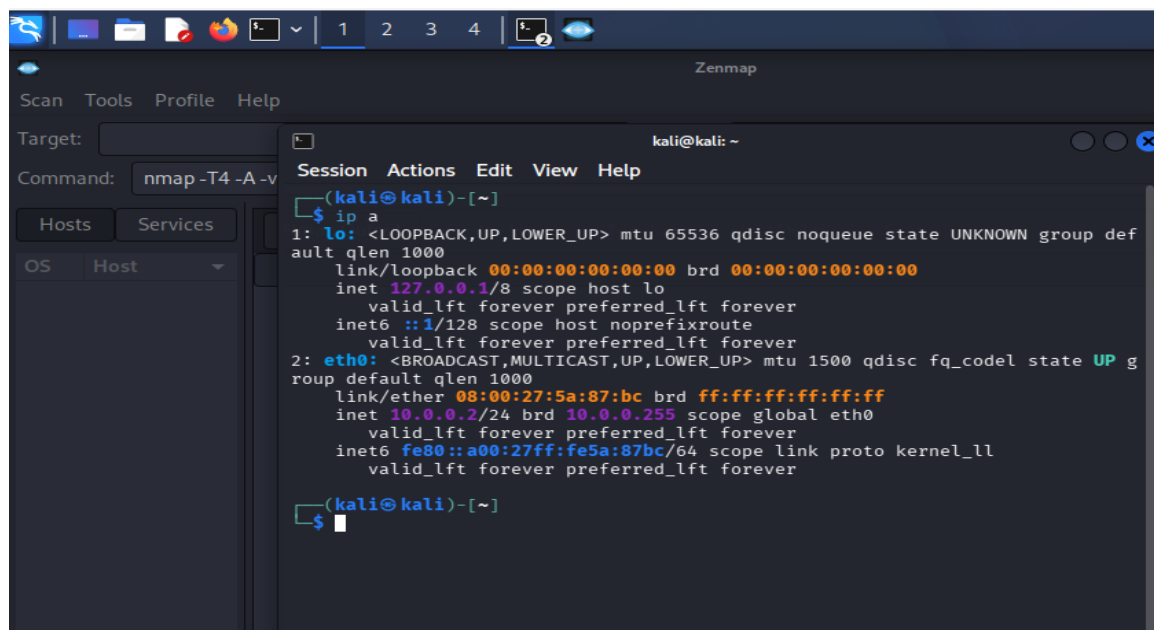
52:54:00:12:35:00

Observations

The scan confirmed connectivity between the virtual machine and the configured NAT network. Zenmap successfully identified devices responding to network probes and provided visibility into network topology.

Evidence

Figure 3: IP Address Configuration



```
zenmap
Scan Tools Profile Help
Target:
Command: nmap -T4 -A -v
Hosts Services
OS Host
Session Actions Edit View Help
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.2/24 brd 10.0.0.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe5a:87bc/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
(kali@kali)-[~]
$
```

Figure 4: Zenmap Scan Results

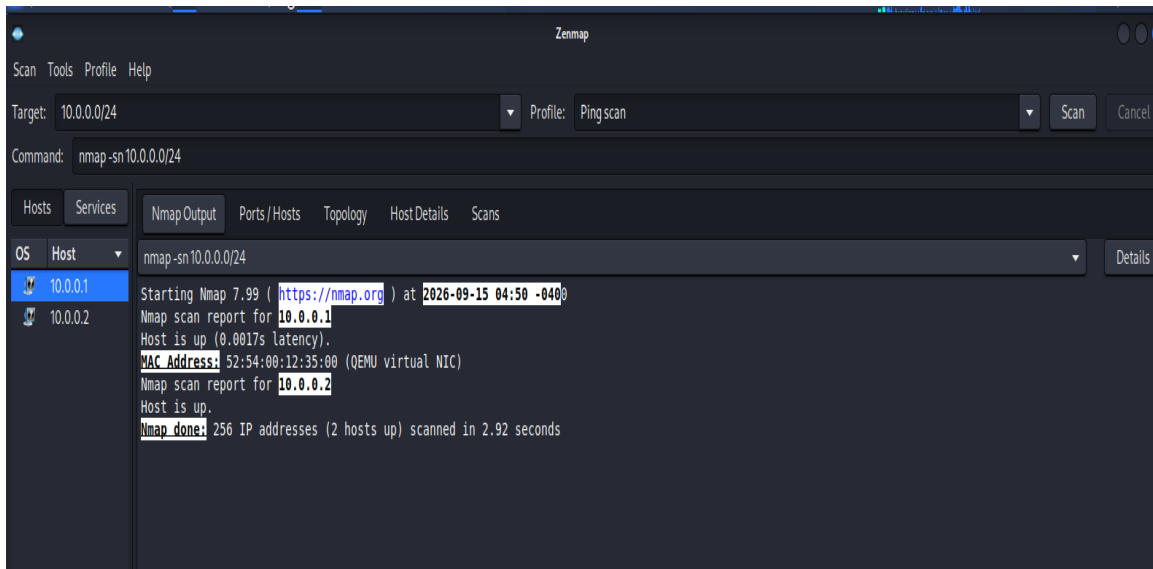
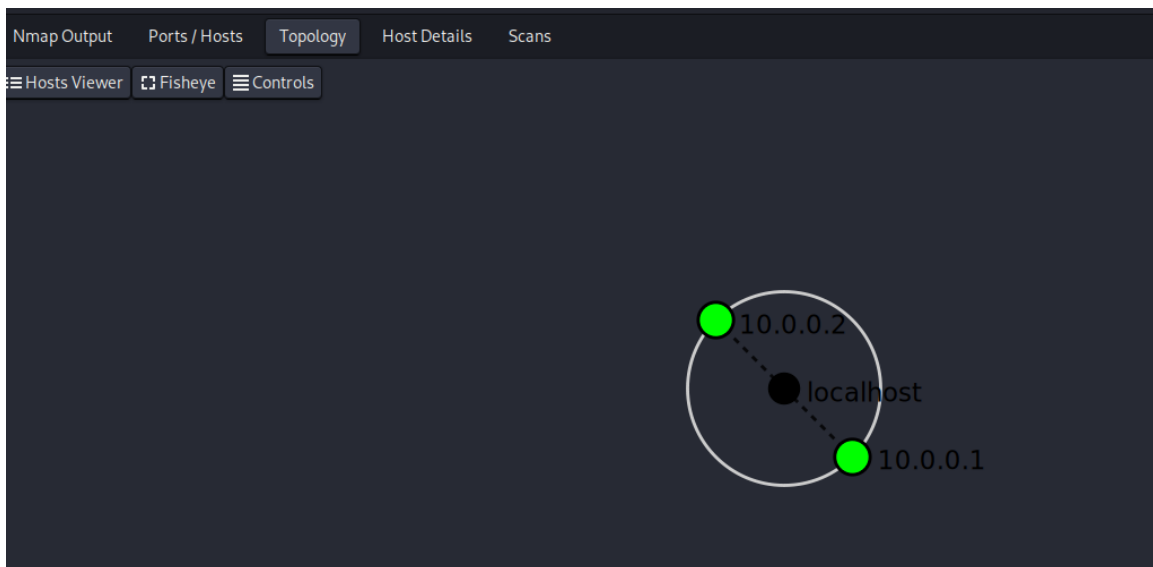


Figure 5: Network Topology



Risk Analysis

Information disclosed through public sources can assist attackers during the reconnaissance phase of a cyberattack.

Publicly exposed subdomains, services, and infrastructure information can provide valuable intelligence that may be used to identify potential attack targets.

Regular monitoring of exposed assets is essential to reduce organizational risk.

Recommendations

- Monitor publicly accessible assets regularly.
 - Review exposed domains and services.
 - Implement asset management practices.
 - Conduct periodic security assessments.
 - Limit unnecessary information exposure where possible.
-

Lessons Learned

During this project I learned:

- Footprinting techniques
 - Open Source Intelligence (OSINT)
 - Passive Reconnaissance
 - Host Discovery
 - Network Enumeration
 - Use of theHarvester
 - Use of Zenmap
 - Nmap Scanning Techniques
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Conclusion

The Week 2 activities provided practical exposure to reconnaissance and network scanning methodologies used in cybersecurity assessments.

TheHarvester demonstrated how publicly available information can be collected from open sources, while Zenmap provided visibility into active devices within the lab environment.

The project strengthened my understanding of information gathering, network discovery, and cybersecurity reconnaissance processes.

-End-
